Predicting Volkswagen Returns from Supply-Chain Signals: A Machine Learning Study

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1. Abstract

This project asks whether the daily returns of Volkswagen's suppliers and two market proxies can forecast the future returns of Volkswagen preferred shares (VOW3.DE) at horizons of 1, 5 and 20 trading days. We build a leak-free pipeline that downloads and cleans 2,797 trading days of data (2015 to 2026), turns each of 10 series into 5 lagged daily returns (50 features), and compares five regressors (Linear Regression, Elastic Net, Decision Tree, XGBoost, LightGBM) against a zero-return random-walk baseline. Models are evaluated with an expanding walk-forward scheme with a label embargo and a final untouched holdout of about one year. A threshold-based long/short strategy with 0.1% transaction costs is then backtested.

The main result is negative but informative. On the holdout, no model beats the zero baseline in RMSE and out-of-sample R-squared is slightly negative at every horizon. Some threshold strategies were profitable (for example +23.97% for the Decision Tree at h = 5 and +21.30% for Linear Regression at h = 1), but they made only 8 to 23 trades during one severe VW decline (Buy and Hold lost 28.61%), so they show capital preservation in one regime rather than a robust trading edge. Inspection of the saved models shows that the final Elastic Net shrinks every coefficient to zero and therefore predicts only the historical mean return.

2. Introduction and Motivation

The Efficient Market Hypothesis (EMH) states that prices instantly reflect all public information, which implies that past returns of related stocks should not help predict a stock's future return. Industrial ecosystems may behave differently. Carmakers depend on suppliers for components, and economic shocks can reach the carmaker with a delay. As the industry moved away from strict Just-in-Time production, the health of suppliers has been proposed as a possible lead-lag indicator for original equipment manufacturers (OEMs).

We formalise this as a supervised learning problem. At the close of day t we observe a feature vector X_t of lagged returns, and we want a function g such that the expected h-day forward return of VW, E[y(t,t+h) | F_t], is approximated by g(X_t). The project follows the framework of R. M. Glawion, "Quantitative Trading with Machine Learning" (Stanford CS229), and extends it with stricter data hygiene, a leakage audit and an untouched holdout.

Objectives:

- Build an end-to-end pipeline that never uses future information.
- Compare five machine learning regressors against a random-walk benchmark.
- Convert forecasts into trading decisions and backtest them with transaction costs.
- Report honestly what works, what does not, and why.

3. Data

3.1 Asset universe

Daily adjusted closing prices were downloaded from Yahoo Finance with auto_adjust=True, from January 2015 to October 2026. After cleaning, the common sample contains 2,797 synchronized trading days (2015-10-09 to 2026-10-09).

Group	Series
Target	Volkswagen preferred shares (VOW3.DE, Xetra)
Suppliers	Continental (CON.DE), Schaeffler (SHA0.DE), Valeo (FR.PA), Infineon (IFX.DE), BASF (BAS.DE), Michelin (ML.PA), ElringKlinger (ZIL2.DE)
Market proxies	DAX 40 index (^GDAXI), EUR/USD exchange rate (EURUSD=X)

3.2 Data cleaning protocol

- **IPO cutoff.** Series whose first price is later than 2016-01-04 are excluded instead of being filled with invented pre-listing returns. Knorr-Bremse (KBX.DE, first price 2018-10-12) was dropped for this reason. Schaeffler's October 2015 listing defines the start of the common sample.

- **Calendar alignment.** All series are aligned to the VW Frankfurt trading calendar. Gaps caused by holidays on other exchanges (for example Euronext Paris) are forward-filled for at most 5 days. This is calendar alignment, not synthetic data, and no zero returns are pre-filled.
- **Asynchronous closes.** EUR/USD trades around the clock and its daily close is recorded after Frankfurt closes. Its same-day return would not be known at VW's close, so EUR/USD features start at lag 1.

VW's price history shows a major rally in 2021, a sharp fall in early 2020 and a decline through 2026. Its daily returns have fat tails (large excess kurtosis), which matters for squared-error models because a few extreme days dominate RMSE.

4. Feature Engineering and Targets

For each of the 10 retained series k, the daily return is r(k,t) = P(k,t) / P(k,t-1) - 1. Each series contributes 5 lags (lag 0 to lag 4), except EUR/USD which contributes lags 1 to 5. This gives exactly 10 x 5 = 50 features, named such as VOW3_DE_lag0, CON_DE_lag2 and EURUSD_X_lag3.

The target for horizon h is the forward return y(t,t+h) = P(t+h) / P(t) - 1, with h in {1, 5, 20} trading days. Returns rather than prices are used because they are comparable across assets and much closer to stationary. Lags let the models test whether supplier moves on previous days lead VW.

5. Models

The models were chosen to cover a spectrum from "learns nothing" to "very flexible", because daily returns contain very little signal and extra flexibility mostly fits noise.

Model	How it works	Settings searched (from the code)	Why it is useful here
Zero baseline	Always predicts a 0% return	none	Random-walk benchmark; all skill is measured against it

Model	How it works	Settings searched (from the code)	Why it is useful here
Linear Regression	StandardScaler, then OLS with one weight per feature	none	Simplest lead-lag test; shows the cost of overfitting 50 correlated inputs
Elastic Net	StandardScaler, then linear model with L1 and L2 penalties	alpha_rel in {0.001, 0.003, 0.01, 0.03, 0.1, 0.3} and l1_ratio in {0.1, 0.5, 0.9}; alpha = alpha_rel x std(y)	Handles correlated suppliers and removes useless lags
Decision Tree	Recursive binary splits; each leaf predicts the mean return of its samples	max_depth in {2, 3, 5}, min_samples_leaf in {20, 50}	Captures thresholds and interactions; readable rules
XGBoost	Boosted trees added sequentially, each correcting the previous residuals	150 trees, learning_rate 0.03, subsample 0.8, colsample_bytree 0.8, reg_lambda 10; max_depth in {2, 3}	Tests for non-linear structure with strong regularisation
LightGBM	Gradient boosting with leaf-wise trees	150 trees, learning_rate 0.03, subsample 0.8, colsample_bytree 0.8, reg_lambda 10, min_child_samples 30; num_leaves in {4, 8}	Same purpose, faster, different tree growth

5.1 Elastic Net in detail

Elastic Net minimises

(1 / 2n) * ||y - X theta||^2 + alpha * (rho * ||theta||_1 + ((1 - rho) / 2) * ||theta||_2^2)

where rho is the l1_ratio. The L1 term can set coefficients exactly to zero, which performs feature selection among the 50 lagged returns. The L2 term shrinks correlated coefficients together, which suits suppliers whose returns move similarly (for example Continental and Schaeffler). Because the penalty depends on feature scale, a StandardScaler is placed inside the pipeline and fitted on training data only.

The notebook sets alpha = alpha_rel x std(y) so that the penalty strength is comparable across horizons, since the target's spread grows from about 0.019 at h = 1 to about 0.076 at h = 20. In development windows at h = 1 about 47.5% of the coefficients were zero on average.

5.2 Trees and boosting

A single Decision Tree with depth 3 and at least 20 samples per leaf has at most 8 leaves, so it is deliberately coarse. XGBoost and LightGBM use small trees, a slow learning rate (0.03), strong L2 regularisation (reg_lambda = 10) and random subsampling of rows and columns. These choices are the standard defence against overfitting noisy financial data.

6. Validation Methodology

Standard k-fold cross-validation shuffles future observations into the training set and is invalid for time series. We use an expanding walk-forward scheme instead.

Component	Setting
Initial training window	1,260 trading days (about 5 years), expanding
Validation window	252 days (about 1 year), used only to choose hyperparameters
Test block	21 days (about 1 month), rolled forward; 61 windows
Embargo	A gap of h days between train, validation and test
Holdout	Final about 1 year (271 days at h = 1, 2025-09-15 to 2026-10-08), never used for any choice

In each window the best hyperparameters are selected on the validation block, the model is refit on training plus validation data, and it predicts the next 21 test days. For h = 1 the first window uses training rows 0 to 1259, validation rows 1261 to 1512 and test rows 1514 to 1534.

Why the embargo. A label at row i is built from the price at row i + h. Without a gap, the last training labels would overlap the first test days and leak information. The code asserts that every label used for training or validation ends before the first test row.

Leakage audit. Observations after each test block were corrupted with Gaussian noise and predictions stayed identical (tolerance 1e-12), showing no backward leakage. The notebook also checks feature-target alignment, identical dates across models and the baseline, backtest unit tests and reproducibility of the saved final model, and deletes its outputs if any check fails.

7. Trading Strategy

At the close of day t the model forecasts the h-day return y_hat. If y_hat is above +c the strategy goes long (+1); if below -c it goes short (-1); otherwise it keeps its current position. The position earns the next trading day's return and is updated daily. Capital starts at 1,000 EUR and a proportional cost of 0.1% is charged on entries, position flips (2 units) and the final liquidation.

The threshold c is chosen from {0, 0.25%, 0.5%, 1%} for each model and horizon by maximising development-period profit at 0.1% cost. The holdout is then run exactly once with that c. Strategies are compared with Buy and Hold and a Cash (no position) benchmark.

8. Results

8.1 Prediction accuracy on the holdout (h = 1)

RMSE ratio is the model's RMSE divided by the zero baseline's RMSE (below 1.00 would mean skill).

Model	RMSE	OOS R-squared	Directional accuracy
Zero baseline	0.0189	0.000	not applicable
Elastic Net	0.0189	-0.003	46.5%
LightGBM	0.0190	-0.007	56.8%
Decision Tree	0.0191	-0.023	46.5%
XGBoost	0.0192	-0.027	50.2%
Linear Regression	0.0196	-0.077	45.8%

At longer horizons Elastic Net remained closest to the baseline: RMSE 0.0396 versus 0.0393 at h = 5, and 0.0770 versus 0.0756 at h = 20, with R-squared of -0.016 and -0.037. Across all models and horizons the holdout RMSE ratio is at or above 1.00, so no model improves on a forecast of zero.

8.2 Trading results on the holdout (0.1% cost, start 1,000 EUR)

Strategy	h	c	Net profit	Trades	Sharpe	Max drawdown
Buy and Hold VW	1	none	-28.61%	1	-0.89	-34.58%
Buy and Hold VW	5	none	-28.04%	1	-0.89	-33.72%
Buy and Hold VW	20	none	-14.48%	1	-0.38	-31.30%
Cash	none	none	0.00%	0	0.00	0.00%
Linear Regression	1	1.00%	+21.30%	8	0.77	-24.73%
Decision Tree	5	0.50%	+23.97%	12	0.86	-19.69%
LightGBM	5	0.50%	+14.56%	23	0.59	-17.49%
Elastic Net	1	0.50%	0.00% (cash)	0	0.00	0.00%
Elastic Net	20	0.50%	-7.88%	6	-0.13	-26.29%

9. Analysis

More flexible models fit more noise. At h = 1 the ordering of holdout RMSE follows model flexibility: the heavily shrunk Elastic Net is nearest the baseline, trees and boosting are slightly worse, and unpenalised Linear Regression is worst (about 4% above baseline). This matches the expectation for a low signal-to-noise problem.

Directional accuracy is not skill by itself. LightGBM's 56.8% hit rate at h = 1 is modestly above 50%, but with about 271 holdout days and several models and horizons tested it is weak evidence, and it did not reduce squared error.

The Elastic Net result needs careful reading. Inspecting the saved final model (alpha_rel = 0.1, l1_ratio = 0.9, alpha about 0.00206) shows that all 50 coefficients are exactly zero. The model outputs only its intercept, 0.000297, the average daily VW return in the training data. This explains why its RMSE ratio is about 1.001, why it never crossed the 0.5% threshold on the holdout (0 trades, cash), and why the "latest prediction" of +0.0297% with a LONG signal is simply that constant rather than a forecast from supplier data. Elastic Net won the development-period selection because it learned almost

nothing, which is the correct behaviour when the signal is weak. A second saved Elastic Net with a weaker penalty (alpha = 0.0005) keeps 28 of 50 coefficients, each of order 0.001 or smaller.

Profitable thresholds reflect regime, not edge. Linear Regression, which had the worst RMSE, produced +21.30%. Profits came from 8 to 23 trades inside one year in which VW fell sharply, so they mostly show that avoiding the stock beat holding it. The choice of c on development data also adds selection risk.

10. Limitations

- **One holdout window.** About one year, in one market regime.
- **Very few trades.** 8 to 23 trades cannot support statistical claims about profitability.
- **Multiple comparisons.** Five models, three horizons and four thresholds increase the chance of a lucky result.
- **Single stock and region.** Only VW and European suppliers are studied.
- **Daily frequency.** Costs, slippage and short-selling constraints are simplified to a flat 0.1%.
- **Limited tuning.** Boosting grids contain two values each, so stronger or weaker settings were not explored.
- **Features are lagged returns only.** No fundamentals, volatility measures, news or order-book data are used.

11. Conclusion

Supplier and market returns can be incorporated into a rigorous, leak-aware forecasting framework for Volkswagen, but in this data none of the five models improves on a zero-return forecast in holdout RMSE. The evidence is consistent with a nearly efficient market at daily horizons: regularisation removes most of the apparent signal, and unregularised or very flexible models do worse out of sample. The positive trading results during a bear market are encouraging but too few and too regime-specific to claim a robust strategy.

Suggested future work (our proposals, not results of this project):

- Test across several years and market cycles with multiple holdout periods.
- Add volatility, volume or fundamental features and longer lags.
- Evaluate with statistical tests on forecast accuracy and with bootstrapped confidence intervals on Sharpe ratios.

- Extend to other OEM and supplier groups to check whether the finding generalises.

Appendix A. Reproducibility

- Random seeds are fixed (seed 42) and the data end date is fixed at 2026-10-10.
- Software versions: scikit-learn 1.9.1, xgboost 3.4.1, lightgbm 4.7.0, pandas 2.2.3, numpy 2.2.1, yfinance 1.7.0.
- Saved artefacts: final_model.joblib, feature_columns.json, model_info.json, pipeline_summary.json, plus results CSVs (prediction_metrics, trading_development_grid, trading_holdout, walk_forward_windows, data_policy_report, latest_prediction) and figures (holdout_equity, rmse_vs_zero, vw_price_and_returns).
- Loading the models with a different scikit-learn version produces a warning; use the versions above.

Appendix B. Key definitions

- **RMSE ratio:** model RMSE divided by zero-baseline RMSE; above 1.00 means worse than predicting zero.
- **OOS R-squared:** 1 minus the model's squared error divided by the baseline's; negative means worse than the baseline.
- **Sharpe ratio:** mean strategy return divided by its standard deviation, annualised.
- **Embargo:** the h-day gap that keeps overlapping labels out of the test set.